

19L820-Project Work 2

BATCH NO:19

Deep Learning based Channel Estimation For OFDM Wireless Communication

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PROBLEM STATEMENT

Channel State Information (CSI) is critical for the performance of **OFDM-based wireless communication systems**. Traditional estimation methods, **Least Squares (LS)** and **Linear Minimum Mean Squared Error (LMMSE)**, face significant challenges:

- **Sensitivity to Noise:** Traditional methods often suffer from severe performance degradation in **low SNR** environments.
- **Statistical Dependency:** LMMSE requires **prior knowledge of channel statistics**, which are often unavailable in real-world scenarios.
- **Limitations of Existing DL Models:** CNNs offer improvements, they often fail to capture **long-range time-frequency correlations**

By these limitations developing a software-based framework that cascades an **Initial Denoising Network (IDN)** with a **Transformer-based model**. This approach aims to suppress pilot noise and exploit **global correlations** across the OFDM grid to reconstruct high-fidelity CSI without requiring prior channel statistics.

LITERATURE SURVEY 1

Deep-Learning Based Channel Estimation for OFDM Wireless Communications

Authors: Guoda Tian, Xuesong Cai, Tian Zhou, Weinan Wang, Fredrik Tufvesson

Journal: IEEE International Workshop on Signal Processing Advances in Wireless Communications (SPAWC)

Year: 2022

Work Done:

- Proposed a deep learning–based channel estimation pipeline for OFDM systems to estimate full CSI from limited pilot symbols **without increasing pilot power** or density.
- Introduced an **Initial Denoise Network (IDN)** and a **Resolution Enhancement Network (REN)** using fully connected layers to improve pilot SNR and recover high-resolution CSI.
- Achieved significant **performance gains** (10–15 dB NMSE improvement) over LMMSE and ChannelNet, especially at low SNR, with a compact model suitable for embedded systems.

Limitations:

- Evaluated only for **SISO OFDM** systems.
- Requires **supervised training** with large labeled datasets.

LITERATURE SURVEY 2

Deep Learning-Based Channel Estimation with Transformer

Authors: J. Guo, et al.

Journal: IEEE Access (2021)

Year: 2021

Work Done:

- **Emphasized the use of Self-Attention** to effectively capture the complex, non-local time-frequency correlations inherent in wireless channels.
- Demonstrated that the Transformer model outperforms CNN-based models by achieving a **lower NMSE (Normalized Mean Square Error)**
- Confirmed that the Transformer architecture is uniquely suited for processing sequential and global grid data by integrating **Positional Encoding** to preserve the time-frequency structure.

Limitations:

- The implementation is **computationally expensive** due to the high complexity of the self-attention mechanism, potentially increasing training time compared to simpler CNN/MLP models.
- Performance is highly dependent on the **quality of the positional encoding** scheme used.

LITERATURE SURVEY 3

Denoising Generalization Performance of Channel Estimation in Multipath Time-Varying OFDM Systems

Author: Yinying Li, Xin Bian, Mingqi Li

Journal: Sensors (MDPI)

Year: 2023

Work Done:

- Designed a deep network with Denoising CNN (DnCNN) to **suppress AWGN** from pilot-based CSI matrices.
- Modeled the channel frequency response as a 2D grid and applied **residual learning** to refine the initial LS/LMMSE estimate.
- Achieved **lower NMSE** and better generalization across varying SNR and Doppler conditions.

Limitation:

- Depends on the accuracy of the initial LS/LMMSE estimate; **large errors propagate** into the denoiser.
- Requires channel/noise statistics implicitly learned from large **labeled datasets**.
- **Increased computational and memory cost** due to deep CNN and residual blocks.

LITERATURE SURVEY 4

ChannelNet: A Deep Neural Network for Channel Estimation in OFDM Systems

Authors: Ye Yuan, Guanding Yu, Hailin Zhang, Yonghui Li

Journal: IEEE Wireless Communications Letters

Year: 2019

Work Done:

- Proposed **ChannelNet**, a CNN-based deep learning framework for OFDM channel estimation.
- Designed a two-stage architecture consisting of a **super-resolution CNN** and an **image restoration CNN** to reconstruct high-quality CSI from sparse pilot signals.
- Treated channel estimation as an image recovery problem by representing CSI as a 2D time–frequency grid.
- Demonstrated **improved NMSE performance** compared to traditional LS and MMSE estimation methods.

Limitations:

- CNN architecture mainly captures **local spatial features** and struggles to learn long-range time–frequency dependencies.
- Requires **high computational resources** for training deeper CNN layers.
- Performance **depends heavily** on pilot distribution and training **dataset** diversity.

LITERATURE SURVEY 5

A DFT-based Low Complexity LMMSE Channel Estimation Technique for OFDM Systems

Authors: Jyoti Prasanna Patra, Bibhuti Bhusan Pradhan, Poonam Singh

Journal: Journal of Telecommunications and Information Technology (JTIT)

Year: 2022

Work Done

- **Reduced the computational complexity** of conventional LMMSE by exploiting DFT properties and structured correlation.
- Achieved near-optimal NMSE/BER performance compared to standard LMMSE and significantly better performance than LS.
- Demonstrated that LMMSE remains a strong and commonly used baseline in OFDM channel estimation.

Limitation

- Still **requires channel statistics / covariance information**, which may not always be accurately available in practical systems.
- **Performance** depends on the **assumed** channel correlation model.
- Although **complexity** is reduced, it is still higher than LS and may be **heavy** for **very low-power devices**.

OBJECTIVE

- To develop a **software-based OFDM** channel estimation framework using deep learning techniques
- To **analyze the limitations** of traditional channel estimation methods under **noisy** and **low-SNR** conditions
- To design and implement a **pilot denoising network** to improve the quality of received channel information
- To apply a **Transformer-based model** for learning global time–frequency channel correlations
- To achieve **improved channel estimation** accuracy compared to conventional and existing **deep learning** approaches
- To evaluate system performance using error metrics such as **Mean Squared Error (MSE)** across different SNR levels

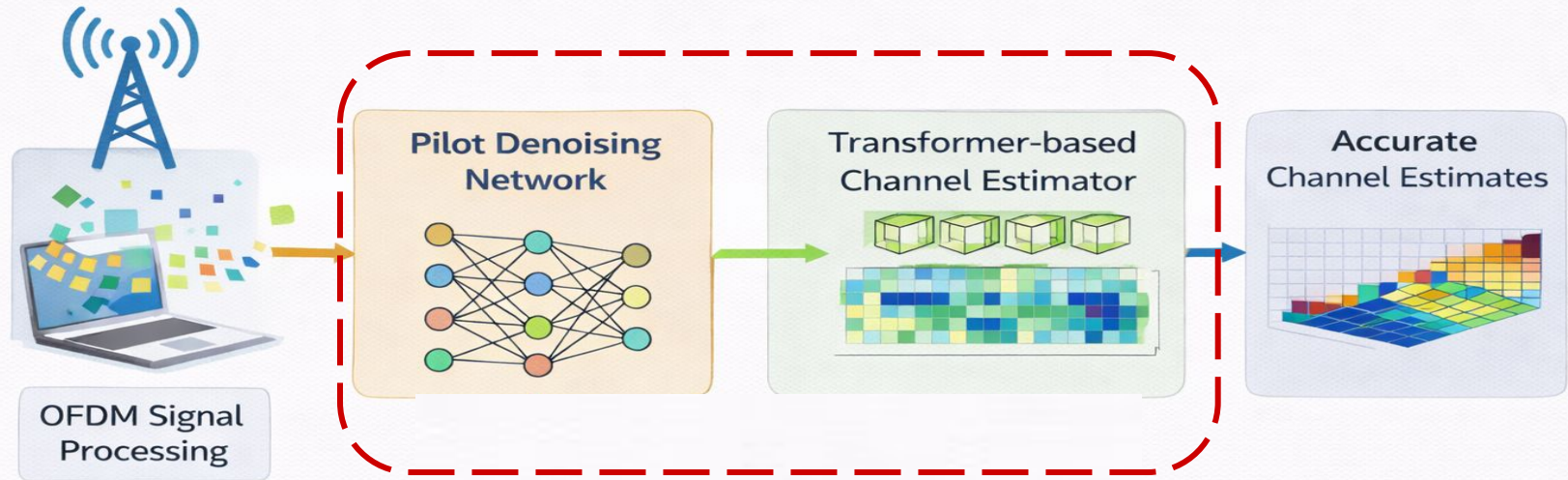
SCOPE OF THE PROJECT

- **Software-Defined Deep Learning Framework:** Engineering a comprehensive software-based estimation pipeline specifically designed for high-fidelity CSI recovery in OFDM wireless systems.
- **Realistic Channel Environment Simulation:** Modeling complex wireless signal conditions utilizing Rayleigh Fading and Additive White Gaussian Noise (AWGN) to analyze estimation robustness.
- **Transformer Architecture:** Implementing a Transformer-based model to exploit multi-head self-attention mechanisms for learning global time-frequency correlations across the resource grid.
- **Quantitative Performance Evaluation:** Assessing system accuracy through standardized MSE and NMSE metrics across a dynamic range of Signal-to-Noise Ratio (SNR) levels.
- **Standardized Benchmarking:** Conducting a comparative analysis to validate the proposed model against conventional Least Squares (LS) and Linear Minimum Mean Squared Error (LMMSE) techniques.
- **Simulation-Based Research Boundaries:** Focus is restricted to high-performance simulation-based modeling using Python, PyTorch, and Google Colab to validate architectural efficiency rather than physical hardware implementation.

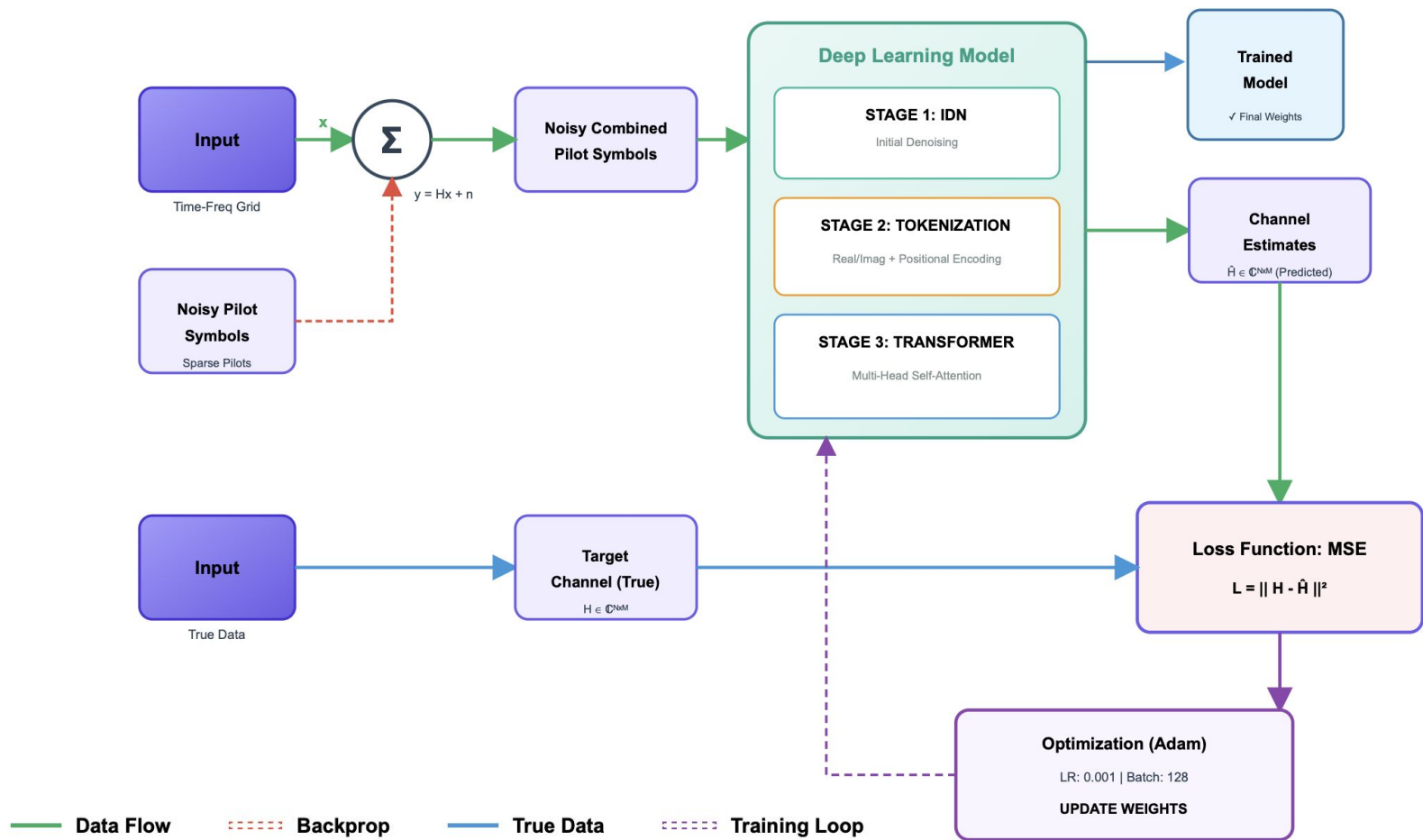
PROPOSED METHODOLOGY

- A software-based deep learning framework is developed for accurate OFDM channel estimation using pilot symbols only.
- The received pilot **Channel State Information (CSI)**, corrupted by noise, is first processed using an **Initial Denoising Network (IDN)** implemented as a fully connected neural network to suppress noise and enhance pilot quality.
- The denoised pilot CSI is then converted into a sequence of tokens by separating real and imaginary components and appending time–frequency positional information.
- A **Transformer-based channel estimation model** is employed to capture long-range correlations across subcarriers and OFDM symbols using self-attention mechanisms.
- The Transformer reconstructs the complete CSI of the OFDM time-frequency grid from the limited pilot information.
- The entire model is trained using a supervised learning approach with **Mean Squared Error (MSE) loss**, without requiring prior channel statistics or hardware measurements.
- **Performance** is evaluated under **multiple Signal-to-Noise Ratio (SNR)** conditions and compared with conventional estimation techniques to demonstrate robustness, particularly in low-SNR scenarios.

PROPOSED METHODOLOGY



BLOCK DIAGRAM OF PROJECT FLOW



TOOLS USED

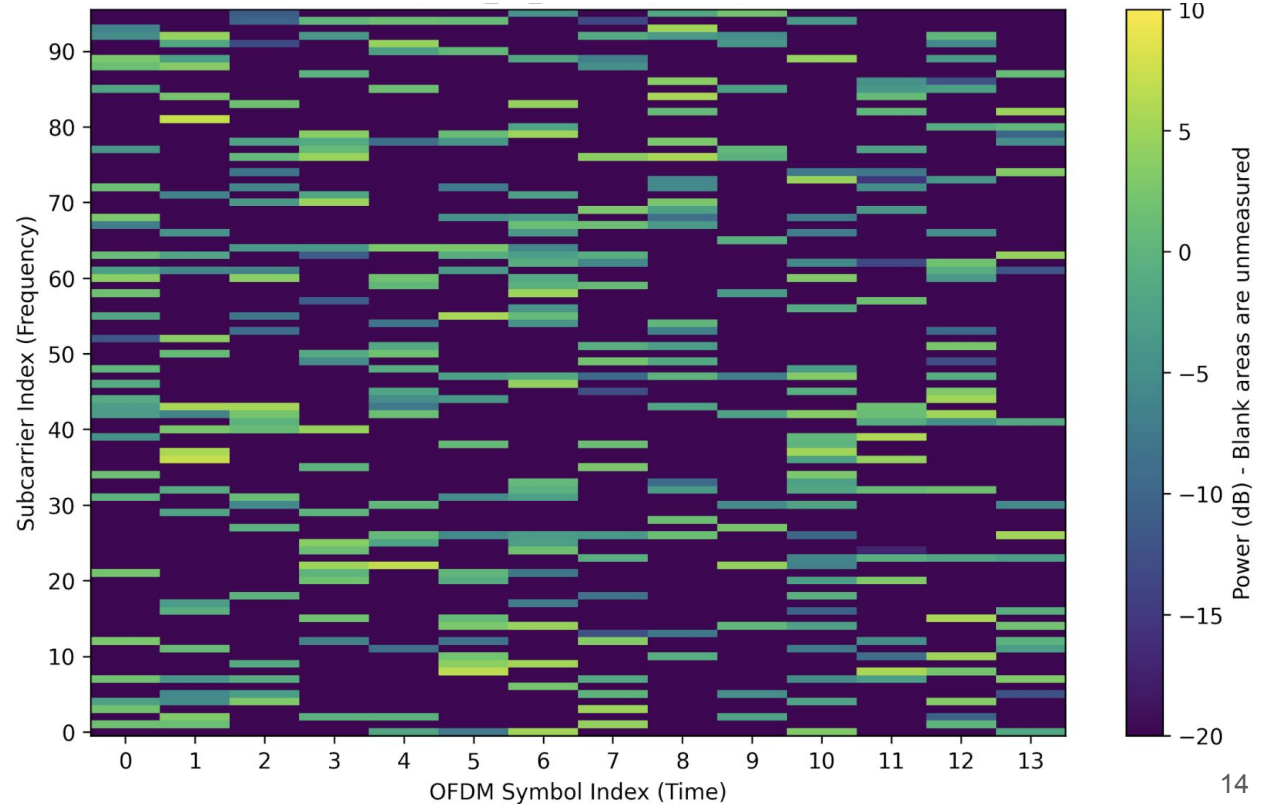
- **Python** – Core programming language for OFDM simulation and deep learning implementation
- **Google Colab** – Cloud-based platform for model training and evaluation with GPU support
- **PyTorch / TensorFlow** – Deep learning framework for implementing the denoising network and Transformer model
- **NumPy & SciPy** – Numerical computation and signal processing operations
- **Matplotlib** – Visualization of performance metrics such as MSE and BER
- **Dataset** – <https://research.ece.ncsu.edu/ai5gchallenge/#datasets>

DATASET

Parameters of the dataset:

Parameter	Value
Channel Model	Rayleigh Fading
Multi-Path Profile	Exponential Power Decay
No. of Subcarriers (N_{SC})	96
No. of OFDM Symbols (N_{SYM})	14
Total Resource Elements	$96 \times 14 = 1344$
Pilot Pattern Type	Scattered
Pilot Density	1/4 (25% of grid)
Modulation	4-QAM / 16-QAM
Performance Metric	NMSE (Normalized MSE)

2D Representation of Data:



PROGRESS:CNN TECHNIQUES

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CNN MODEL COMPARISON TABLE

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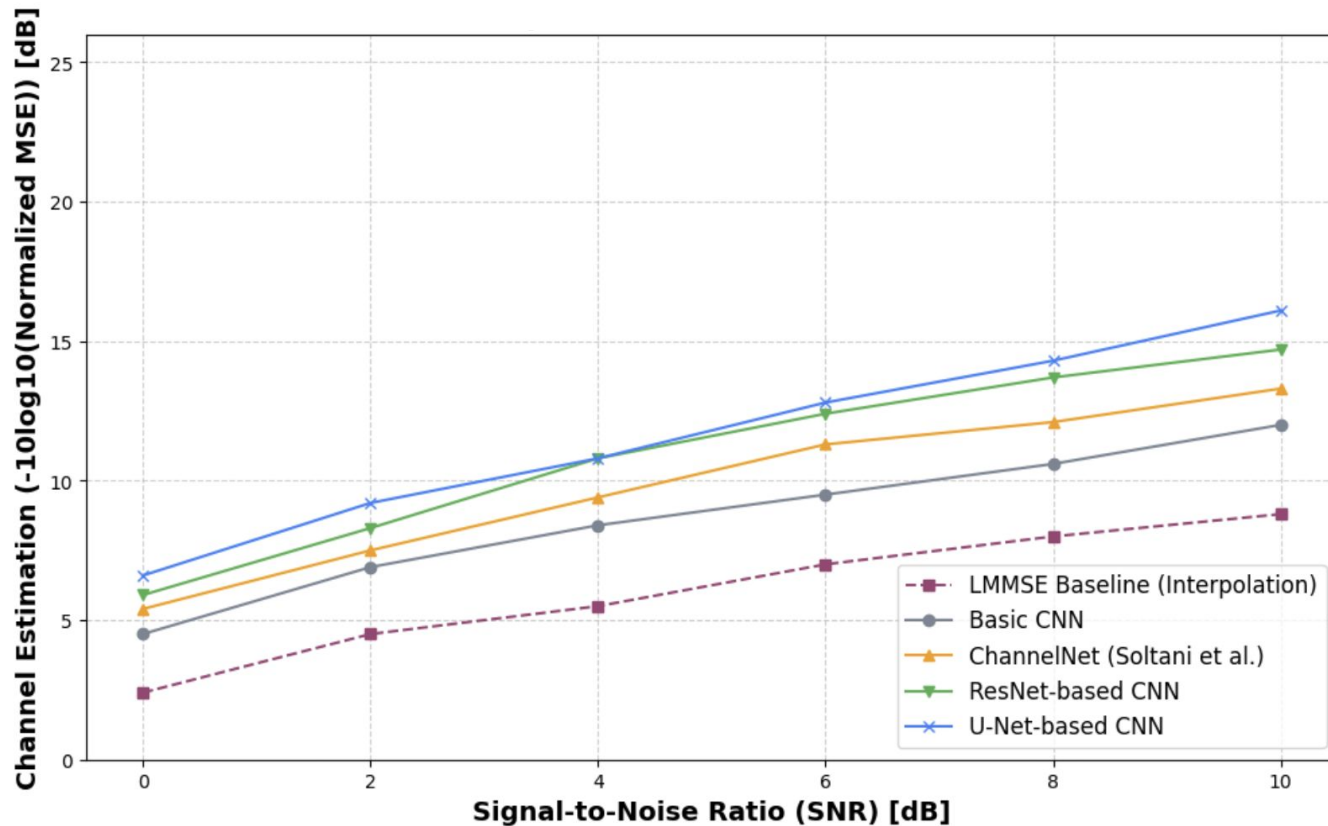
Model	Parameters	Depth/Type	Key Connections/Mechanism
basic_cnn	185,922	Simple 6-layer CNN (Encoder-Interpolation-Decoder)	None (Sequential)
channelnet	1,038,724	SRN (3 blocks) + IRN (5 blocks)	Residual (within blocks)
resnet	684,194	8x Residual Blocks (Deep Feature Extraction)	Residual (between Conv layers)
unet	1,782,978	Encoder-Decoder (2 levels)	Skip (Concatenation across levels)
attention_cnn	751,234	Encoder-Decoder with SE-style blocks	Channel Attention (Weighting)

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Dataset used:

Parameter	Value	Parameter	Value
Channel Model	Rayleigh Fading	Total Resource Elements	$96 \times 14 = 1344$
Multi-Path Profile	Exponential Power Decay	Pilot Pattern Type	Scattered / Sparse
No. of Subcarriers (N_{SC})	96	Pilot Density	$1/4$ (25% of grid)
No. of OFDM Symbols (N_{SYM})	14	Modulation	16-QAM

CNN TECHNIQUES-PERFORMANCE



PRESENT TECHNIQUES

1. Least Square(dB):

$$\hat{H}_{LS}(k) = \frac{Y(k)}{X_{\text{pilot}}(k)}$$

Estimates the channel by simply **dividing the received pilot signal by the known transmitted pilot symbol**, minimizing the energy of the error in the pilot domain.

2. Linear Minimum Mean Squared Error (LMMSE) (dB):

$$\hat{\mathbf{h}} = \mathbf{C}_{hx} \cdot \mathbf{C}_{xx}^{-1} \cdot \mathbf{x},$$

C_{hx} : Represents the correlation between the true channel and the received signal.
 C_{xx} : Represents the correlation of the received signal with itself.

It is a **statistical linear interpolation** method that uses pre-calculated **covariance matrices** to weight received pilot signals and minimize the mean squared error across the channel grid.

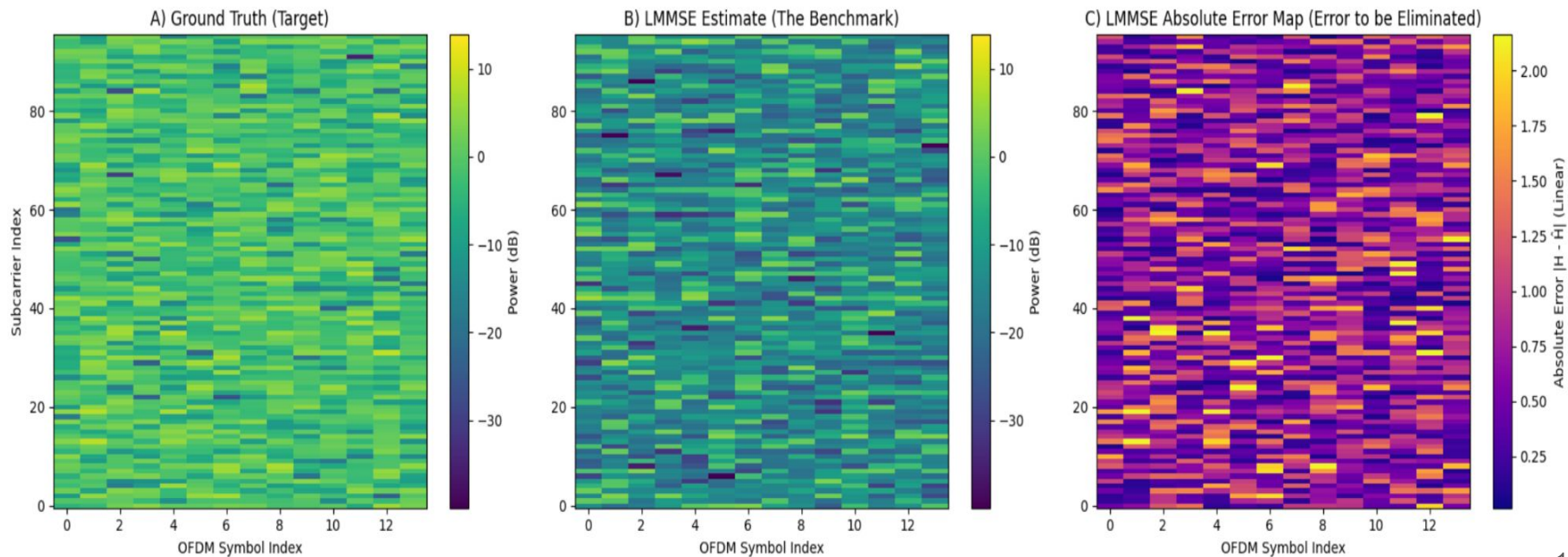
$$\text{NMSE} = 10 \cdot \log_{10} \left(\frac{\mathbb{E}[|H_{\text{True}} - \hat{H}|^2]}{\mathbb{E}[|H_{\text{True}}|^2]} \right)$$

Output:

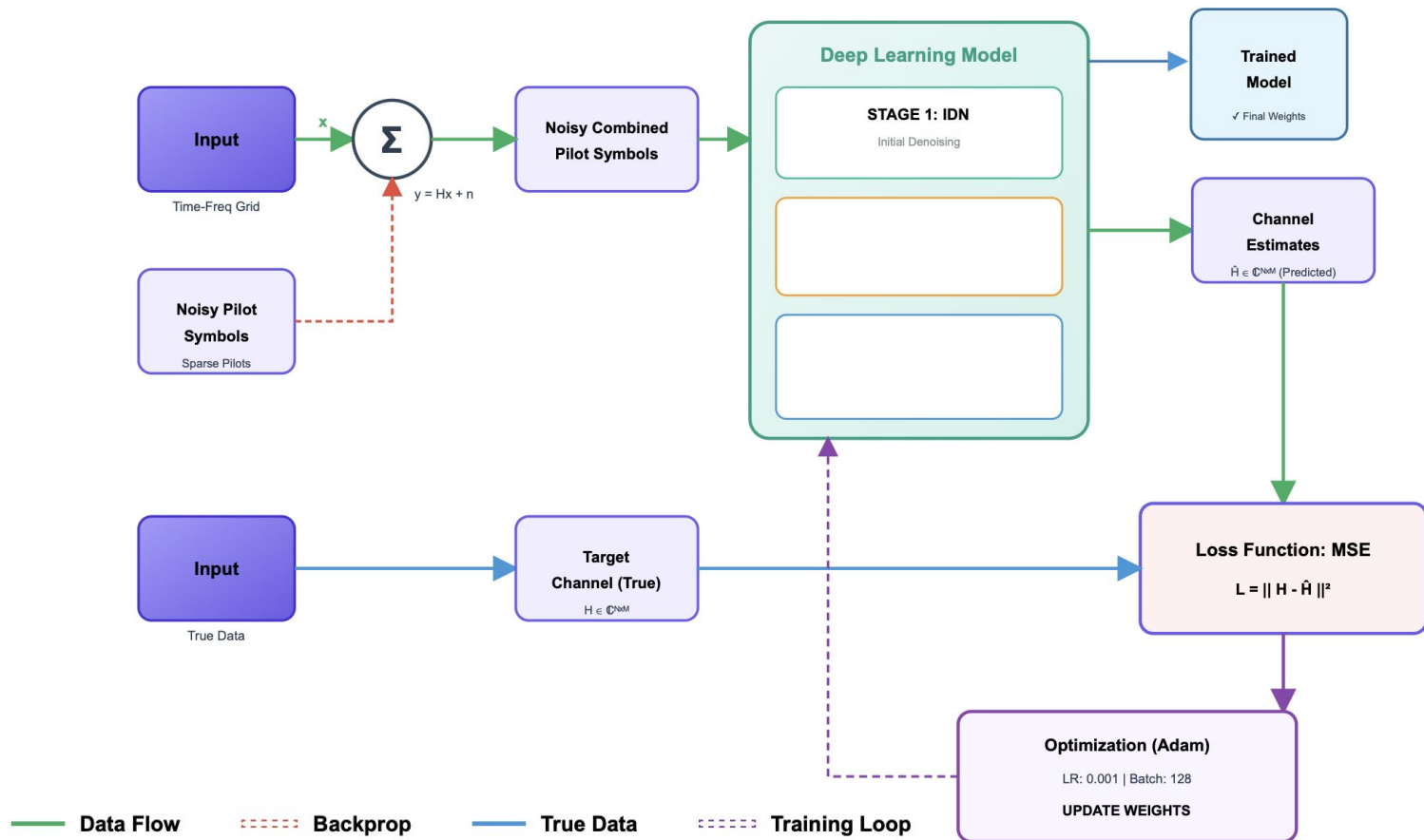
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--- BENCHMARK RESULTS (SNR = 10 dB) ---  
1. LS (Noisy, Sparse Input) NMSE: -1.20 dB  
2. LMMSE (Optimal Traditional Method) NMSE: -1.37 dB  
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LMMSE TECHNIQUE

Channel Estimation LMMSE: -1.37dB



FLOWCHART OF INITIAL DENOISING NETWORK



THEORY OF IDN

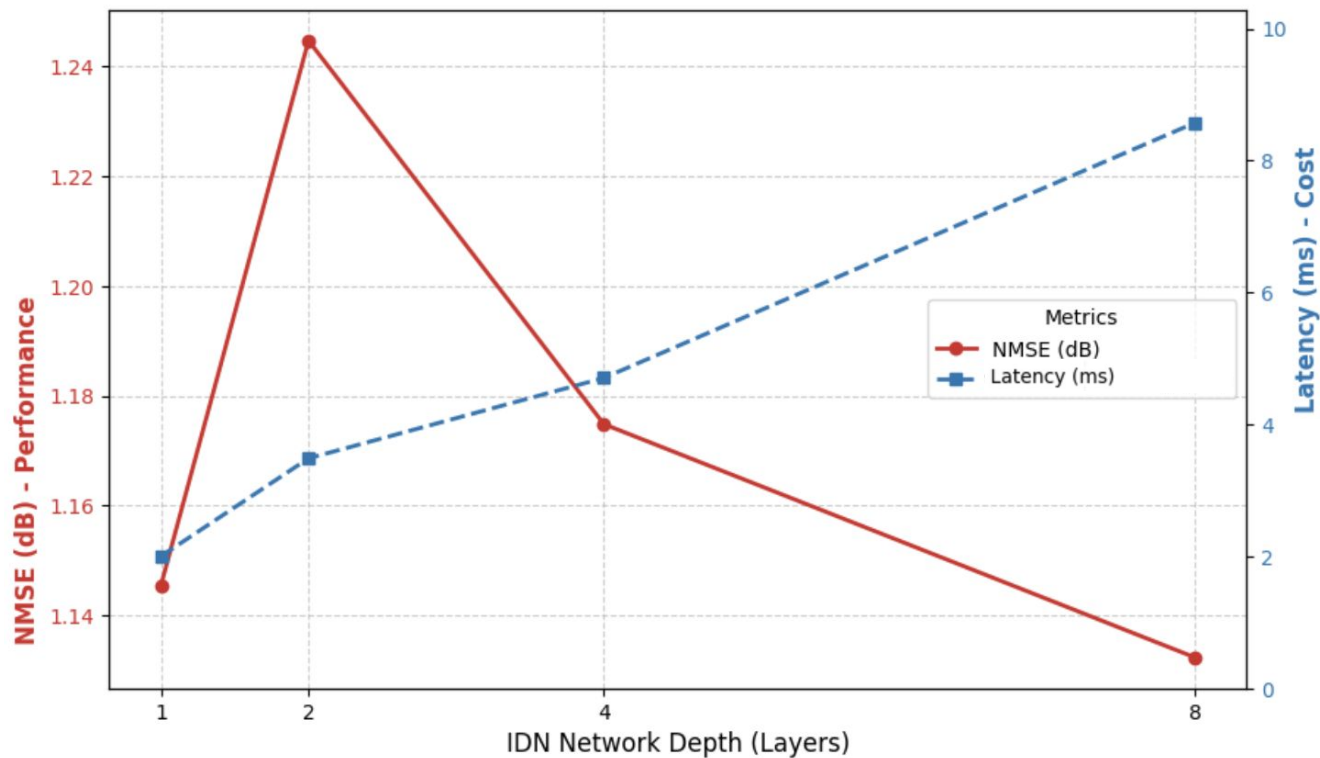
IDN Output Formula : $\hat{\mathbf{H}}_{\text{denoised}} = \mathbf{H}_{LS} - f_{\text{IDN}}(\mathbf{H}_{LS})$

- **$\mathbf{H}_{\text{denoised}}$** : The **Denoised Channel State Information (CSI)**, which is the final output of the network.
- **$\mathbf{H}_{LS}$** : The **Least Squares (LS) Channel Estimate**, which serves as the noisy input to the IDN.
- **$f_{\text{IDN}}(\mathbf{H}_{LS})$** : The **Learned Noise Residual** predicted by the 4-layer dense network to be subtracted from the noisy input

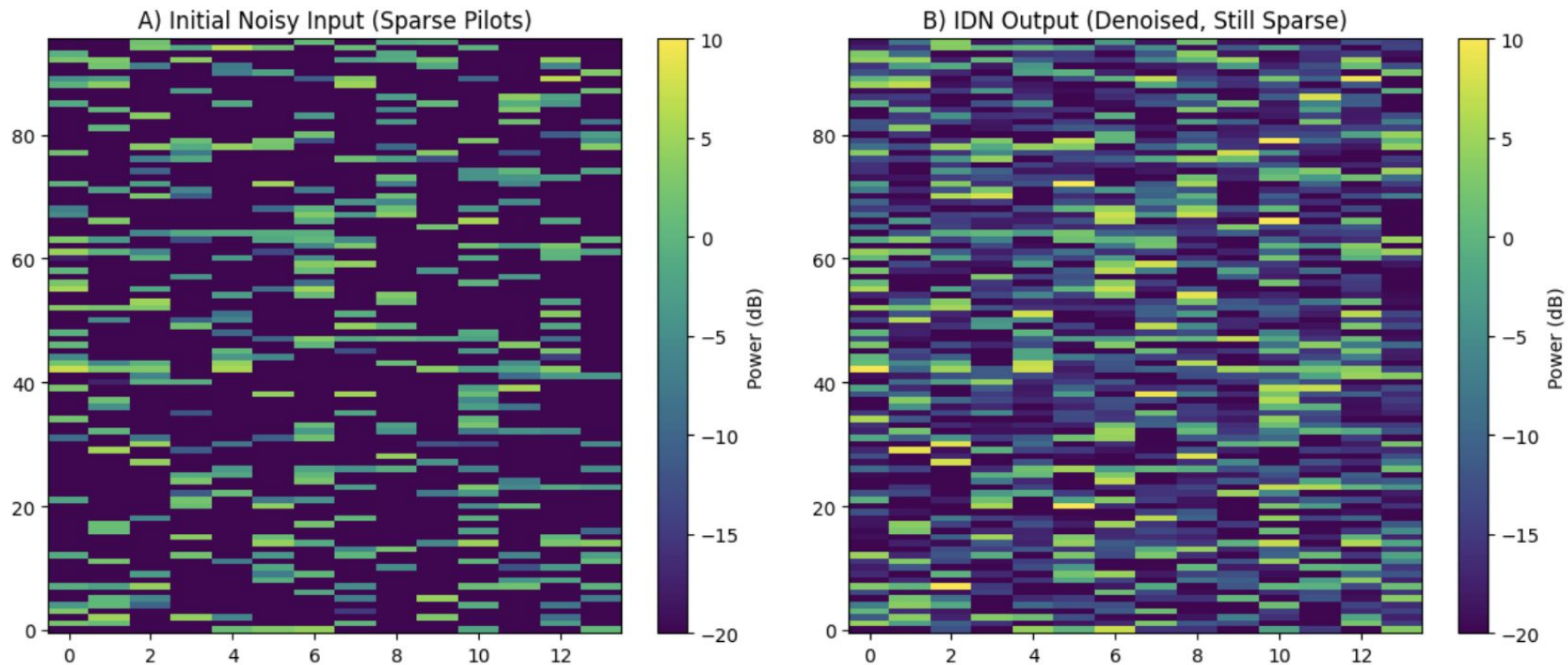
Parameters of the IDN:

Parameter	Value
Layers	4 Dense Layers
Activation	LeakyReLU
Normalization	Batch-Normalisation
Output	$\hat{\mathbf{H}}_{\text{denoised}}$

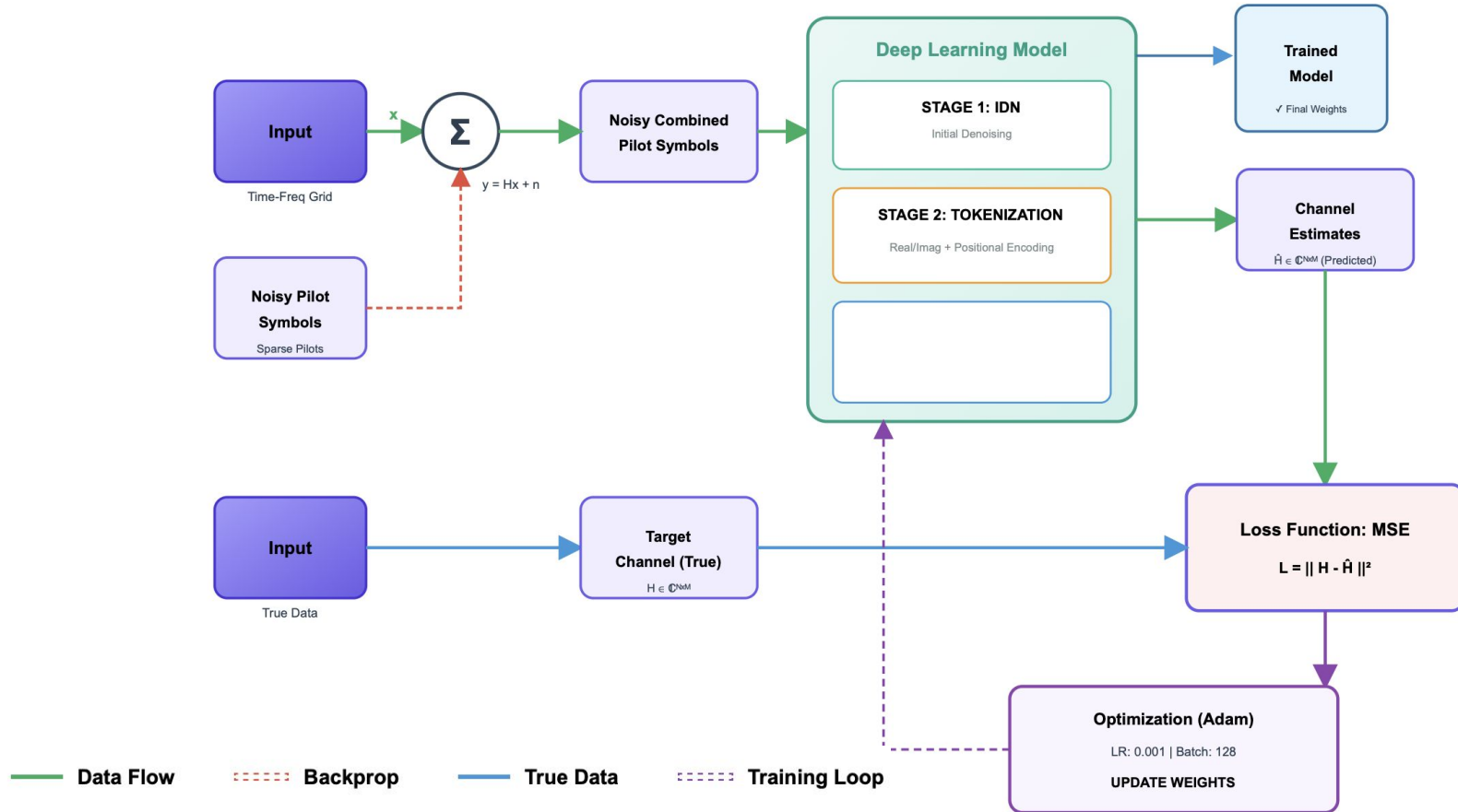
DETERMINATION OF LAYERS IN IDN



INPUT & OUTPUT LAYERS IN IDN



FLOWCHART OF TOKENIZATION



THEORY OF TOKENIZATION

1. Objective

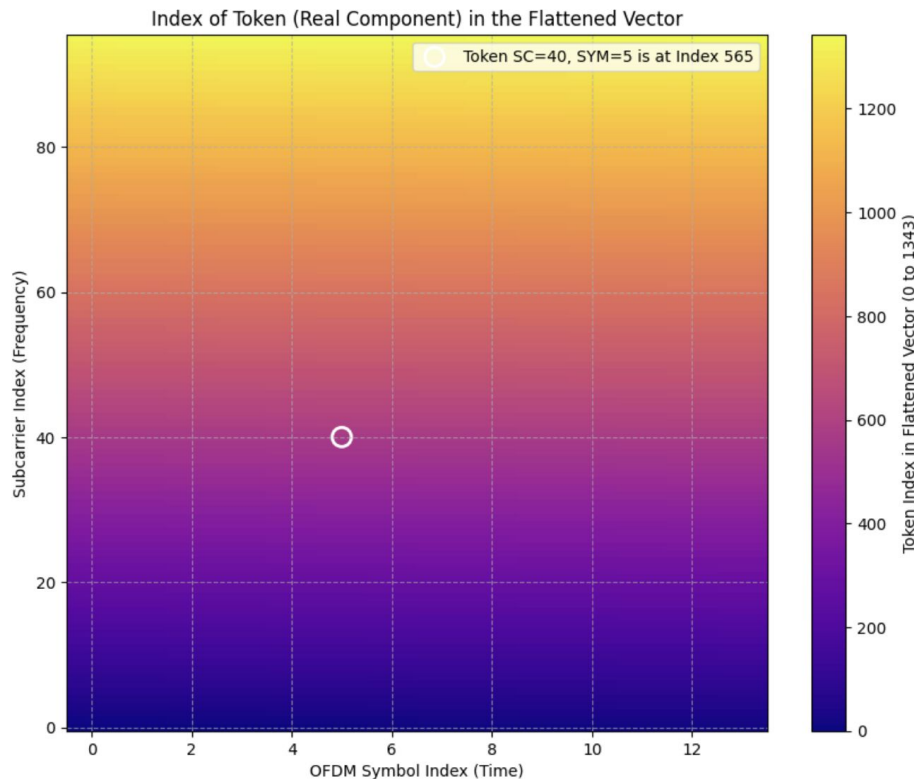
- **Problem:** The Transformer requires **1D sequential data** (like a sentence) but the Channel State Information (**H**) is a **2D grid** (Frequency $\times$ Time).
- **Goal:** Convert the 2D channel grid into a structured 1D sequence **without losing the crucial spatial coordinates**.

2. Process

Step	Concept	Description
A. Serialization	Flattening the Grid	The clean pilot data ($\mathbf{H}^{\text{denoised}}$) is flattened into a single, long vector (length $\text{NSC} \times \text{NSYM} \times 2$).
B. Token Creation	Real/Imag Separation	The complex values are separated into their Real and Imaginary components, forming the core features of the sequence.
C. Positional Encoding	Injecting Location	Append the original Subcarrier and Symbol index to each element. This tells the Transformer the element's exact location on the grid.

$$\text{Token}_k = [\text{Re}(\hat{H}_k), \text{Im}(\hat{H}_k), \text{Pos}_{\text{freq}}, \text{Pos}_{\text{time}}]$$

2D VISUALIZATION OF TOKENIZATION

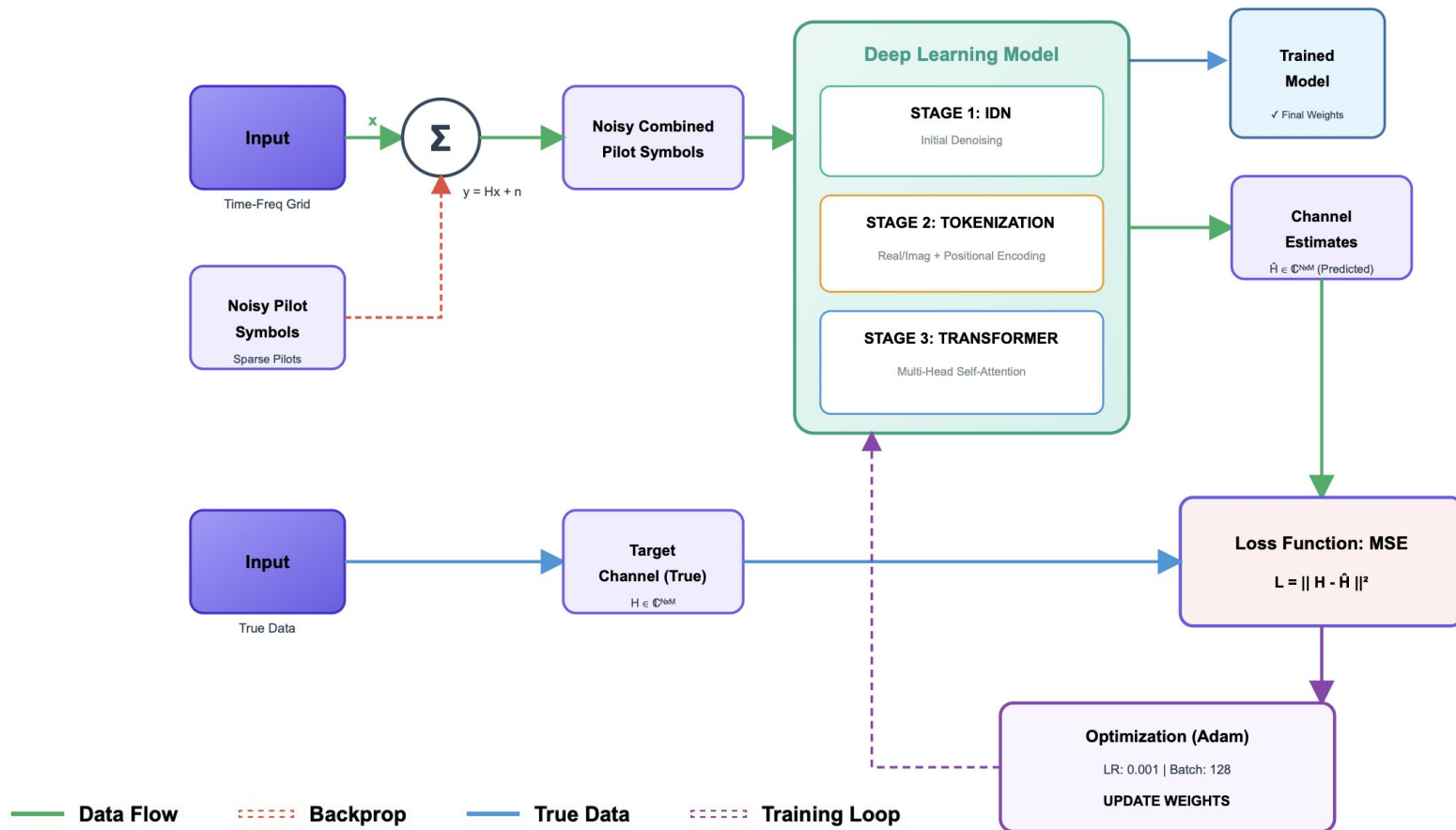


$$\text{Token}_k = [\text{Re}(\hat{H}_k), \text{Im}(\hat{H}_k), \text{Pos}_{\text{freq}}, \text{Pos}_{\text{time}}]$$

The Color Graph

1. **Low Index (Dark Purple/Blue):** These are the elements that come **first** in the sequential vector (e.g., the bottom-left corner).
2. **High Index (Yellow/White):** These are the elements that come **last** in the sequential vector (e.g., the top-right corner).

FLOWCHART OF DEEP LEARNING MODEL



DL MODEL PHASE 1 PARAMETERS

Module	Component	Formula	Parameter Count
IDN	Linear 1	$(2688 \times 1024) + 512$	2,753,536
	BatchNorm 1	2×1024	2,048
	Linear 2	$(1024 \times 512) + 256$	524,800
	BatchNorm 2	2×512	1,024
	Linear 3	$(512 \times 256) + 128$	131,328
	Linear 4	$(256 \times 2688) + 2688$	690,816
Transformer	Linear Embed	$(2688 \times 512) + 512$	1,376,768
	BatchNorm	2×512	1,024
	4x Encoder Layers	$4 \times [d_{\text{model}} \cdot (12 d_{\text{model}} + 13)]$	~12,600,000
	Output Layer	$(512 \times 2688) + 2688$	1,379,456
TOTAL			~19.46 Million

DEEP LEARNING MODEL TRAINING PHASE 1

--- HYPERPARAMETERS ---

BATCH_SIZE = 128

LEARNING_RATE = 0.001

N_EPOCHS = 50

--- STARTING TRAINING ---

Epoch [1/50], Loss: 0.527514

Epoch [2/50], Loss: 0.503611

Epoch [3/50], Loss: 0.495742

Epoch [4/50], Loss: 0.485545

Epoch [43/50], Loss: 0.431157

Epoch [44/50], Loss: 0.430972

Epoch [45/50], Loss: 0.430858

Epoch [46/50], Loss: 0.430710

Epoch [47/50], Loss: 0.430558

Epoch [48/50], Loss: 0.430426

Epoch [49/50], Loss: 0.430358

Epoch [50/50], Loss: 0.430177

--- FINAL PERFORMANCE RESULTS ---

1. LS (Initial) NMSE: -1.20 dB

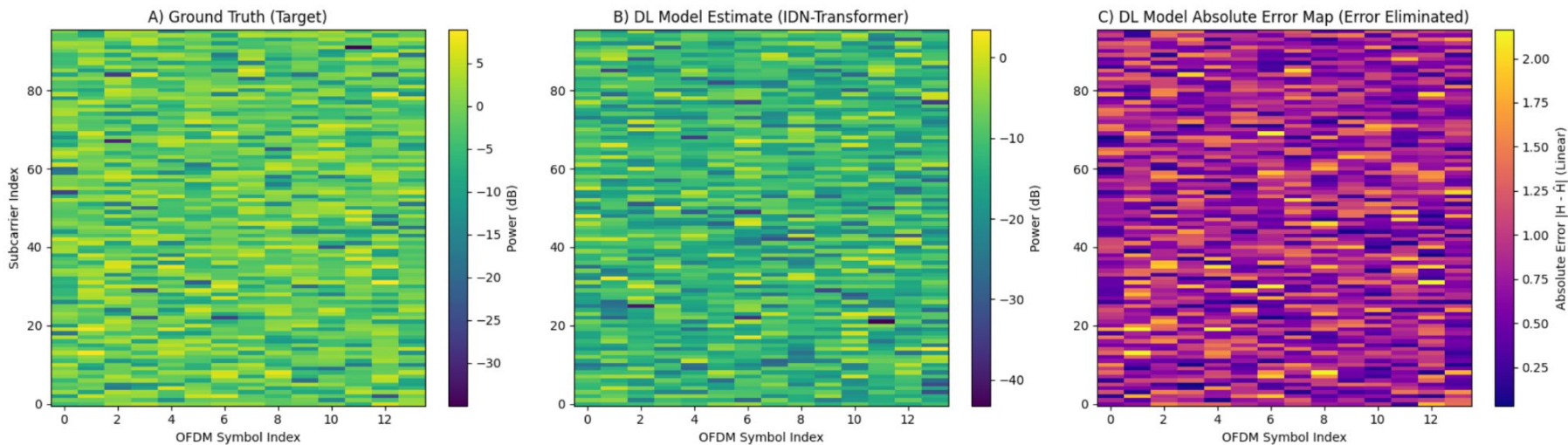
2. LMMSE (Benchmark) NMSE: -1.37 dB

3. DL Model (IDN-Transformer) NMSE: -0.70 dB

DEEP LEARNING MODEL TRAINING PHASE 1

INPUT AND OUTPUT VISUALIZATION

DL Model Channel Estimation Comparison (DL NMSE: -0.70 dB)



Channel Estimation DL Phase 1: -0.70dB

COMPARISON OF DEEP LEARNING MODEL

Inference:

- **Inference Efficiency:** The **0.039 GFLOPs** per inference indicates a lightweight execution phase, essential for deployment on edge devices.
- **Scalability:** The linear increase in **Total Training GFLOPs** from Phase 1 to Phase 2 shows predictable resource consumption during the development cycle.
- **Real-Time Capability:** The consistent **5ms latency** across both phases confirms the model's stability for time-sensitive resource grid processing (96×14).

1. GFLOPs (Inference Complexity):It tells how much work the computer does to process **one single input** (one channel matrix).

2. Total Training GFLOPs (Computational Energy):It tells the total amount of "math" performed during the entire training process to reach the final weights.

Phase	1	2
Metric	50 Epochs	300 Epochs
Total Parameters	1,94,61,472	1,94,61,472
Model Size (MB)	74.2 MB	74.2 MB
GFLOPs (Per Inference)	0.039 GFLOPs	0.039 GFLOPs
Total Training GFLOPs*	58,500 GFLOPs	351,000 GFLOPs
Training Time	8 Minutes	40 Minutes
Latency(approx)	5ms	5ms

DEEP LEARNING MODEL TRAINING PHASE 2

--- HYPERPARAMETERS (INCREASED EPOCHS) ---

BATCH_SIZE = 128

LEARNING_RATE = 0.001

N_EPOCHS = 300

--- STARTING ENHANCED TRAINING (300 Epochs) ---

Epoch [1/300], Loss: 0.528005

Epoch [2/300], Loss: 0.508915

Epoch [3/300], Loss: 0.503364

Epoch [4/300], Loss: 0.493641

Epoch [296/300], Loss: 0.358793

Epoch [297/300], Loss: 0.358903

Epoch [298/300], Loss: 0.358767

Epoch [299/300], Loss: 0.358707

Epoch [300/300], Loss: 0.358711

--- FINAL PERFORMANCE RESULTS ---

1. LS (Initial) NMSE: -1.20 dB

2. LMMSE (Benchmark) NMSE: -1.37 dB

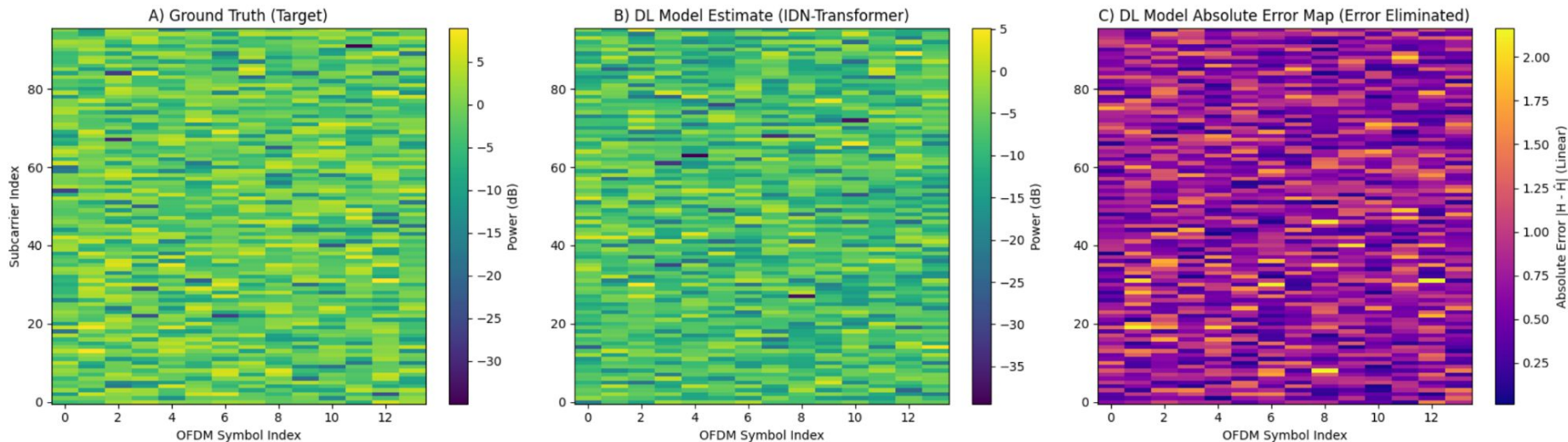
3. DL Model (IDN-Transformer) NMSE: -1.57 dB

DL MODEL PHASE 2 PARAMETERS

Module	Component	Formula	Parameter Count
IDN	Linear 1	$(2688 \times 1024) + 1024$	2,753,536
	BatchNorm 1	2×1024	2,048
	Linear 2	$(1024 \times 512) + 512$	524,800
	BatchNorm 2	2×512	1,024
	Linear 3	$(512 \times 256) + 256$	131,328
	Linear 4	$(256 \times 2688) + 2688$	690,816
Transformer	Linear Embed	$(2688 \times 512) + 512$	1,376,768
	BatchNorm	2×512	1,024
	4x Encoder Layers	$4 \times [d_{model} \cdot (12 d_{model} + 13)]$	~12,600,000
	Output Layer	$(512 \times 2688) + 2688$	1,379,456
TOTAL			~19.46 Million

DEEP LEARNING MODEL TRAINING PHASE 2

DL Model Channel Estimation Comparison (DL NMSE: -1.57 dB)



Channel Estimation DL Phase 2: -1.57dB

RESULT

- **Deep Learning Superiority:** The IDN-Transformer consistently outperforms both the LS and LMMSE mathematical baselines under 10 dB SNR conditions.
- **Effective Noise Mitigation:** The Initial Denoising Network (IDN) stage provides a cleaner input, which is particularly critical for sensitive high-order modulations like **16-QAM**.
- **Global Correlation Capture:** By utilizing a **self-attention mechanism**, the model learns complex, non-local time-frequency correlations across the entire **96×14 resource grid**, overcoming the local limitations of standard CNNs.

Estimation Method	Model Category	NMSE (dB)	Core Strength
Least Squares (LS)	Mathematical Baseline	-1.20	Standard baseline with high noise sensitivity.
LMMSE	Traditional Optimal	-1.37	Traditional optimal linear estimator.
IDN-Transformer	Proposed Model	-1.57	Superior noise suppression & global correlation.

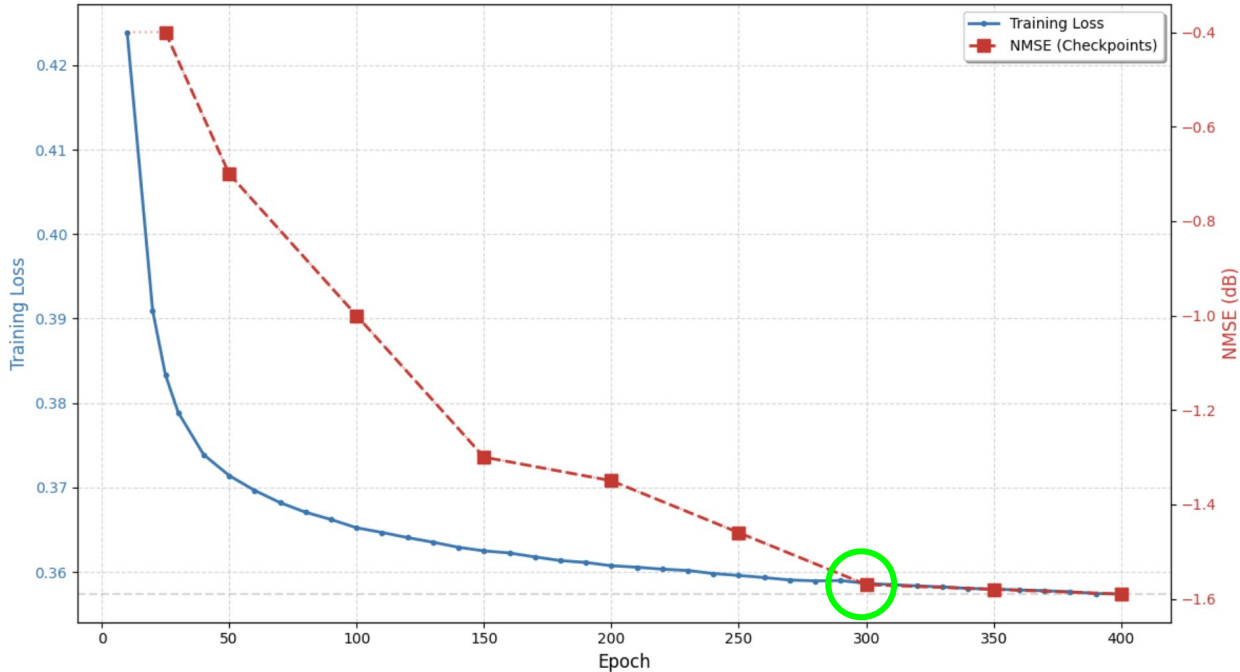
WORKS REMAINING

- 1.DL model training with [IDN]->[Tokenization]->[Transformer] -For 10000 samples
- 2.DL model Optimization and switching 16QAM
- 3.Comparison with Deep Learning model and Linear Minimum Mean Squared Error

PROGRESS

JUSTIFICATION DL MODEL TRAINING PHASE 2

IDN-Transformer: 400 Epoch Convergence Graph



--- FINAL PERFORMANCE RESULTS ---

1. LS (Initial) NMSE: -1.20 dB
2. LMMSE (Benchmark) NMSE: -1.37 dB
3. DL Model (IDN-Transformer) NMSE: -1.57 dB

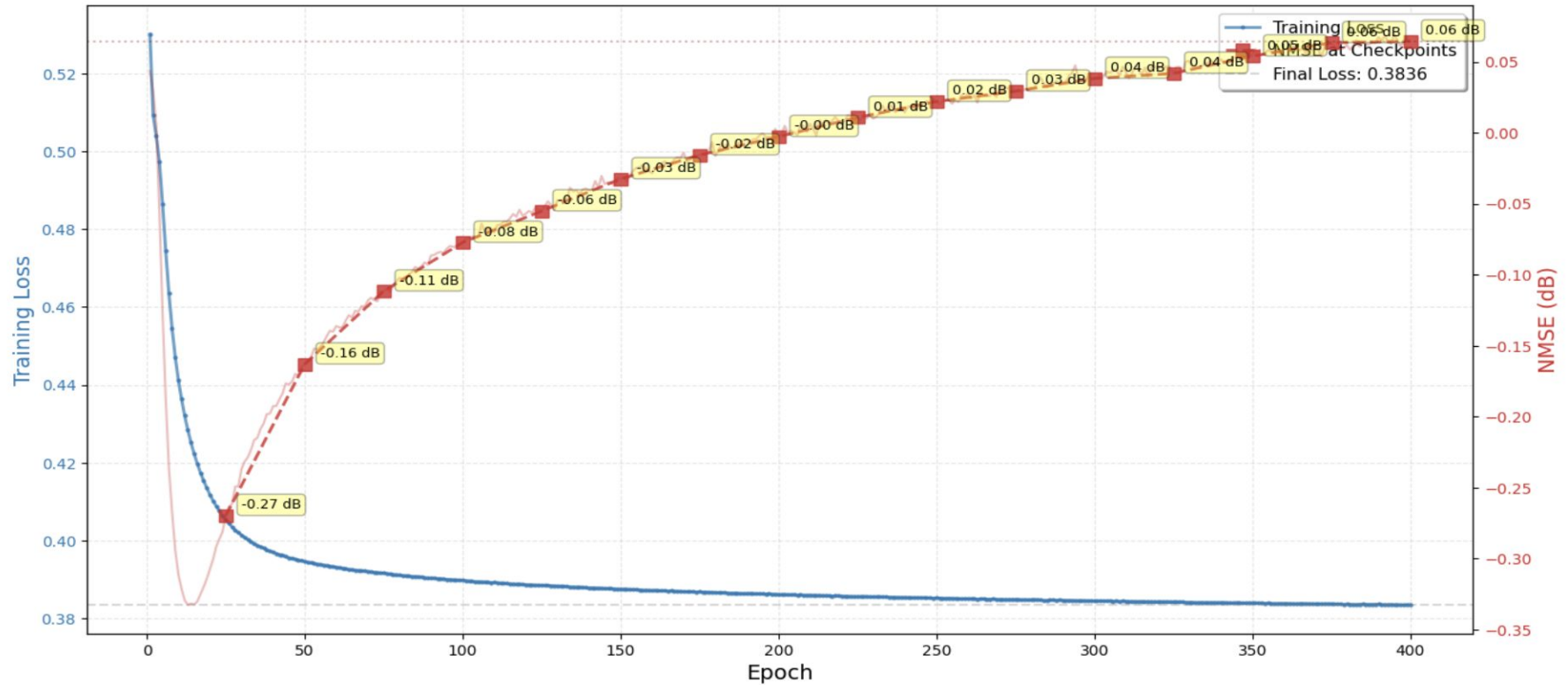
DL MODEL PHASE 3 PARAMETERS REDUCED BY 25%

Module	Component	Formula	Parameter Count
IDN	Linear 1	$(2688 \times 768) + 768$	2,065,152
	BatchNorm 1	2×768	1,536
	Linear 2	$(768 \times 384) + 384$	295,296
	BatchNorm 2	2×384	768
	Linear 3	$(384 \times 192) + 192$	73,920
	Linear 4	$(192 \times 2688) + 2688$	518,784
	Linear Embed	$(2688 \times 384) + 384$	1,032,576
	BatchNorm	2×384	768
Transformer	4x Encoder Layers	$4 \times [384 \times (12 \times 384 + 13)]$	7,086,336
	Output Layer	$(384 \times 2688) + 2688$	1,034,880
	TOTAL		9,759,168(49.9% reduction)

Determination of these specific parameters numbers(768, 384, 192):

- Preserves the 2:1 ratio between consecutive layers
- Maintains enough capacity for channel estimation
- Balanced reduction across all layers
- Keeps dimensions multiples of 64 (good for GPU optimization)

DL MODEL PHASE 3 OUTPUT

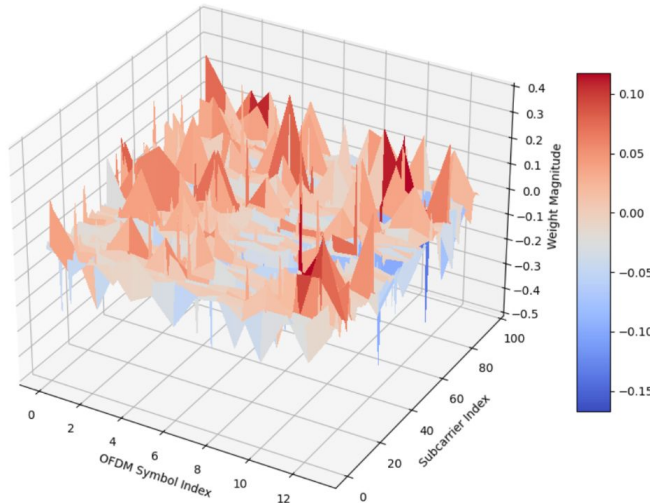


DL MODEL PHASE 3 - JUSTIFICATION

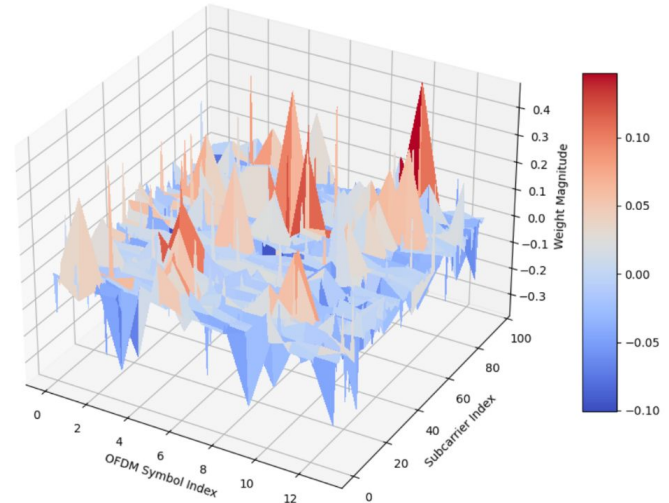
1. **Total Parameter Reduction to 50%**
2. **Flattening the Grid**

Rayleigh fading creates "**peaks**" and "**valleys**" across time and frequency. Adjacent subcarriers and adjacent symbols are highly correlated. By flattening 96 x 14 grid into a single 1D vector and using nn. Linear are completely destroying that 2D geometric relationship. Furthermore, because Transformer's sequence length is forced to 1, **it cannot track how the channel changes over time.**

Real Weight Component (Influence on Power/Gain)



Imaginary Weight Component (Influence on Phase/Shift)



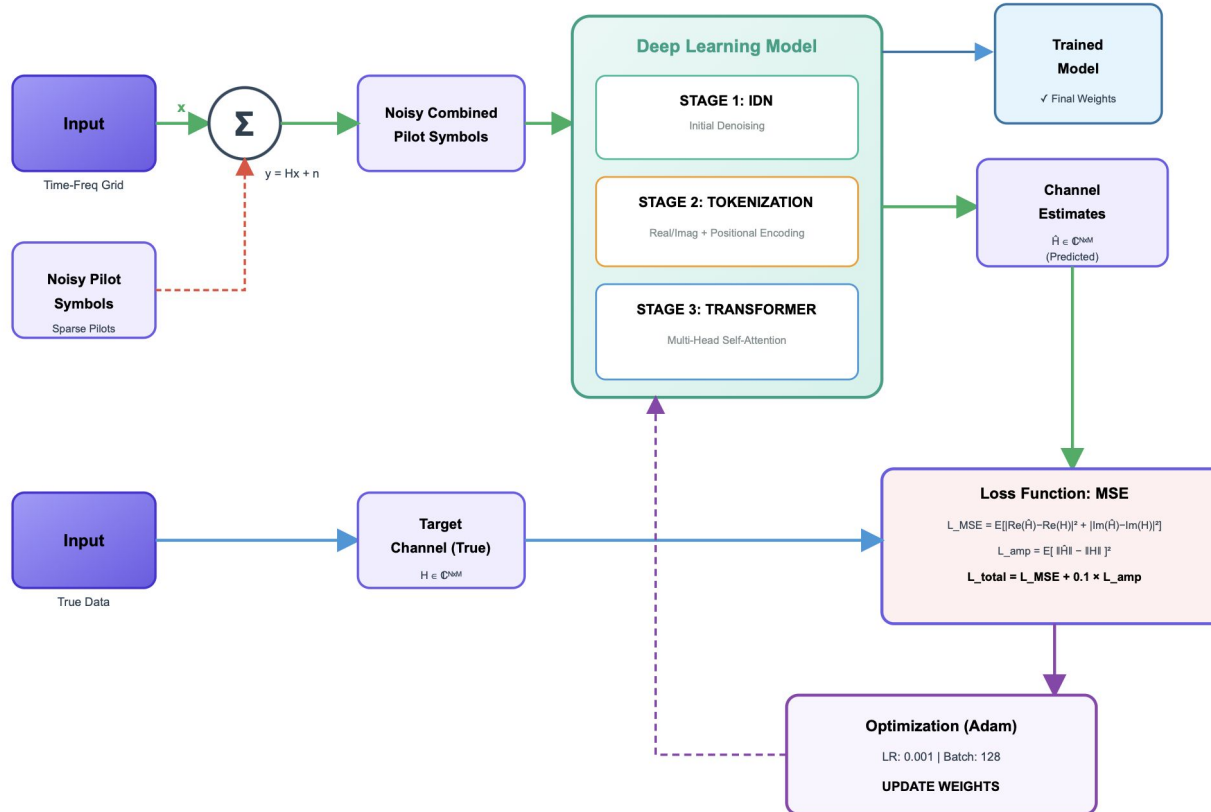
DL MODEL PHASE 3 - VARIETY MODEL

Build and tested model

S No	Total parameters	NSME(dB)	Epoch rate	Training time(seconds)
1.	14,838,064	-0.18 dB	40	490 sec
2.	10,697,312	-0.16 dB	35	251 sec
3.	3,268,866	-1.13 dB	90	226 sec
4.	3,268,866 (with dropout-0.3)	-1.17 dB	60	178 sec

The reduction is based on individual parameter reduced to 15%, 25%, 50% from the original parameter used in DL Model Phase 2

DL MODEL PHASE 3 - OPTIMIZATION



DL MODEL PHASE 3 - OPTIMIZATION

Loss Function Upgradation

Current: Standard MSE

$$L_MSE = E[|Re(\hat{H}) - Re(H)|^2 + |Im(\hat{H}) - Im(H)|^2]$$

- Treats real & imaginary parts independently — no coupling constraint
- A network can minimise MSE while producing wrong complex amplitude $|H|$
- Received signal power depends on $|H|^2$, so amplitude errors directly hurt SNR
- Phase accuracy is implicit only — no explicit geometric penalty in IQ space

The Fix: Amplitude Regularisation

$$L_amp = E[||\hat{H}|| - ||H||]^2 ||\cdot|| = \sqrt{Re^2 + Im^2} \\ \text{(complex magnitude)}$$

$$L_total = L_MSE + 0.1 \times L_amp$$

- Explicitly penalises wrong signal power — ties directly to receiver SNR
- Weight 0.1 keeps amplitude as a regulariser, not the dominant objective
- Analogous to perceptual losses in image super-resolution
- Preserves physically meaningful channel magnitude profiles

DL MODEL TRAINING PHASE 3 WITH OPTIMIZED

Output:

```
Epoch [ 1], Val NMSE: -0.13 dB
Epoch [ 2], Val NMSE: -0.19 dB
Epoch [ 3], Val NMSE: -0.31 dB
Epoch [ 4], Val NMSE: -0.36 dB
Epoch [ 5], Val NMSE: -0.41 dB
```

```
Epoch [46], Val NMSE: -4.34 dB
Epoch [47], Val NMSE: -4.49 dB
Epoch [48], Val NMSE: -4.59 dB
Epoch [49], Val NMSE: -4.68 dB
Epoch [50], Val NMSE: -4.76 dB
```

Output NMSE:

PHASE 1: -0.70dB

PHASE 2: -1.57dB

PHASE 3: -4.76dB

DL MODEL PHASE 3 PARAMETERS

Module	Component	Formula	Parameter Count
IDN	Linear 1	$(1024 \times 512) + 512$	524,800
	BatchNorm 1	2×512	1,024
	Linear 2	$(512 \times 256) + 256$	131,328
	BatchNorm 2	2×256	512
	Linear 3	$(256 \times 128) + 128$	32,896
	Linear 4	$(128 \times 1024) + 1024$	132,096
Transformer	Linear Embed	$(1024 \times 256) + 256$	262,400
	BatchNorm	2×256	512
	2x Encoder Layers	$2 \times [256 \times (12 \times 256 + 13)]$	1,579,520
	Output Layer	$(256 \times 1024) + 1024$	263,168
TOTAL			2,928,256 (~3M params)

WORKS REMAINING

- 1.DL model Optimization and Justification
- 2.DL model training with [IDN]->[Tokenization]->[Transformer] -For 10000 samples
- 3.Comparison with Deep Learning model and Linear Minimum Mean Squared Error

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THANK YOU